

Minjae Kwon

Curriculum Vitae

Dipartimento di Fisica,
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Education

- MAR2015–
FEB2019 **Bachelor of Science (B.S.), Physics**, *Pusan National University*, Busan, Republic of Korea.
- MAR2019–
FEB2022 **Master of Science (M.S.), Physics**, *Pusan National University*, Busan, Republic of Korea.
Thesis: Study on reciprocal scattering of low energy alpha source using silicon pixel detector
Supervisor: In-Kwon Yoo (Pusan National University)
Referee: Sanghoon Lim (Pusan National University), Minjung Kweon (Inha University)
- MAR2022–
AUG2022 **Ph.D Course, Physics**, *Pusan National University*, Busan, Republic of Korea.
- NOV2025–
present **Doctoral Student , Physics**, *University of Turin*, Turin, Italy.
Reasarch Topic: Frontier Readout Technologies for Liquid Detectors

Working

- SEP2022–
OCT2025 **Researcher**, *NSRI*(*Natural Science Research Institute*), *University of Seoul*, Seoul, Republic of Korea.

Interests

Physics

- Exploring QCD phase diagram
- 1st order phase transition between hadronic matter and quark gluon plasma
- Electronics under extreme condition

Engineering & Computation

- Physics Modeling and Simulation
- Silicon Pixel Detector

Experience

DEC2016– FEB2019 **Undergraduated Internship**, *Heavy Ion Physics Experiment Lab. (HIPEX)*, Department of Physics, Pusan National University, Busan, Republic of Korea.

MAR2019– FEB2022 **Student Researcher : Master Course**, *Heavy Ion Physics Experiment Lab. (HIPEX)*, Department of Physics, Pusan National University, Busan, Republic of Korea.

01JUL2019– **Summer Student**, CERN, Meyrin, Switzerland.

24AUG2019 CERN Non-Member State Summer Students 2019

SEP2021– JUN2022 **Local Organizer, Technical Staff**, *SQM2022(The 20th International Conference on Strangeness in Quark Matter)*, Busan, Republic of Korea.

13Jun2022 – 18Jun2022, Paradise Hotel Busan + Zoom Event

(Conference) Webpage Management, Indico+Zoom Broadcasting Management, Responding emails, (Public Lecture) Indico Management, Responding emails, Poster Sending, Document to Schools

<https://sqm2022.pusan.ac.kr>

<https://indico.cern.ch/event/1037821/>

<https://indico.kps.or.kr/event/38>

MAR2022– AUG2022 **Student Researcher : Ph.D Course**, *Heavy Ion Physics Experiment Lab. (HIPEX)*, Department of Physics, Pusan National University, Busan, Republic of Korea.

MAR2019– **Teaching Assistant**, Pusan National University, Pusan, Republic of Korea.

AUG2022 Teaching Assistant for lecture 'General Physics Laboratory' at Spring Semester of 2019, Spring Semester of 2020 and Spring Semester of 2022

SEP2021– AUG2022 **ALICE-ITS On-call**, ALICE, CERN, Meyrin, Switzerland.

2021–2022 **ALICE Underground Guide**, ALICE, CERN, Meyrin, Switzerland.

SEP2022– OCT2025 **Researcher**, *NSRI(Natural Science Research Institute)*, University of Seoul, Seoul, Republic of Korea.

NOV2023– **Indico Translation Team**, *[en→ko-KR] Volunteer-based translation project*, present 10JUL2026 100% translated, 95% contributed.

NOV2023– **InvenioRDM Translation Team**, *[en→ko] Volunteer-based translation project*, present 10JUL2026 100% translated, 100% contributed.

present **Local Organizer**, *Local Organizing Committee, 32nd International Conference on Ultra-relativistic Nucleus-Nucleus Collisions (Quark Matter 2027)*, Jeju, Republic of Korea.

Project

- FEB2017– **Development of Circumstance Monitoring System**, *Pusan National University*,
MAY2017 Busan, Republic of Korea.
System to monitor temperature and humidity was developed with Raspberry Pi, external sensor and Google Spreadsheet for data storing. Data can be served without local server, just with google spreadsheet. It has been running smoothly in clean room for about 2 years. (since May 2017)
- MAY2017– **Mass Chip Test (MCT) and Outer Barrel Hybrid Integrated Circuit Assembly**
MAY2019 **(OB-HIC Assembly)**, *Pusan National University*, Busan, Republic of Korea.
It includes procedures for Mass Chip Test (MCT) and Outer Barrel Hybrid Integrated Circuit Assembly (OB-HIC Assembly).
- JUL2019– **Analysis Of Cosmic Ray Data Taken During The Commissioning Of The**
AUG2019 **New ALICE Inner Tracking System**, *ALICE, CERN*, Meyrin, Switzerland.
Supervised by Magnus Mager and Markus Keil, EP-AID-DT, CERN. The Report is uploaded at CERN CDS : <http://cds.cern.ch/record/2687398>
- JUN2019– **Study on the 3-body Elastic Scattering System**, *Pusan National University*,
FEB2022 Busan, Republic of Korea.
This project is supported by National Research Foundation of Korea (NRF)
- SEP2022– **Development of Neutron Detector using GEM(Gas Electron Multiplier)**, *University of Seoul*, Seoul, Republic of Korea.
FEB2025 This project is supported by National Research Foundation of Korea (NRF)
- FEB2023 **Donghae Project: Donghae Neutrino Observatory**, *University of Seoul+Seoul National University*, Seoul, Republic of Korea.
Designing Measurement System, Pre-design, Visual design/Modeling
- JUN2023– **OpenKIAS “Science with Citizen” [muonpi]**, *University of Seoul*, Seoul, Republic
OCT2025 of Korea.
Collaborated with KIAS. <https://github.com/muonpi-kr>, <https://muonpi.tistory.com>
- OCT2023– **Dual Readout Calorimeter**, *University of Seoul + DRC Collaboration*, Seoul,
AUG2024 Republic of Korea.
Quality measurment of scintillating fiber
- DEC2023 **indico-plugin-payment-tosspayments**, *personal*.
- MAR2024– **Research on Simulation of Radiation Shielding to Maximize Survivaliility in**
DEC2024 **Space**, *University of Seoul*, Seoul, Republic of Korea.
This project is supported by National Research Foundation of Korea (NRF). Commissioned by KIST.
- APR2024– **Development of All-in-one Window-glass for Radiation Shielding, Bullet-**
OCT2025 **Proof and Thermal Insulation**, *University of Seoul*, Seoul, Republic of Korea.
This project is supported by National Research Foundation of Korea (NRF). Collaborated with KIST, KIMS, KAERI.
- FEB2025– **Barrel Imaging Calorimeter**, *University of Seoul + koBIC Collaboration*, Seoul,
OCT2025 Republic of Korea.

Conferences (Oral Presentation, Poster)

- 26OCT2017 **Temperature and Humidity Monitoring System with Raspberry Pi**, *Contest for Undergraduate at KPS Fall Meeting, Gyeongju, Republic of Korea.*
Poster
- 11JUN2018 **OB HIC Production - Pusan/Inha**, *11th ALICE ITS upgrade, MFT and O2 Asian Workshop, Hiroshima, Japan.*
Oral
<https://indico.cern.ch/event/687364/contributions/3036053>
- 04NOV2018 **A Report on Korean activities for the ALICE-ITS upgrade**, *The 7th Asian Triangle Heavy-Ion Conference (ATHIC 2018), Hefei, China.*
Oral
<https://indico.ihep.ac.cn/event/8180/session/35/contribution/181>
- 25OCT2024 **OpenKIAS: Science with Citizen [muonpi]**, *2024 KPS Fall Meeting.*
Oral
- 07DEC2024 **Technical Description of muonpi**, *openKIAS <Science with Citizen> Workshop.*
Oral

Publication

0. Jongsik Eum, **Minjae Kwon**, Sanghyeon Lee, In-Kwon Yoo, "Feasibility Study on the Total Ionizing Dose Effect of the Alice Pixel Detector(ALPIDE) and the Korea Multi-Purpose Accelerator Complex(KOMAC)", *New Phys.: Sae Mulli*, **vol.69**, 283 (2019) (*Co-Author*) [doi:10.3938/NPSM.69.283]
1. **Minjae Kwon**, Magnus Mager, Markus Keil, "Analysis Of Cosmic Ray Data Taken During The Commissioning Of The New ALICE Inner Tracking System)", *CERN Summer Student Programme Report*, CERN CDS : CERN-STUDENTS-Note-2019-128 (2019)
2. **Minjae Kwon**, Yongjun Choi, Jehyeong Kim, Seonguk Park, In-Kwon Yoo, "Double Alpha Scattering Experiment with two $^{241}_{95}\text{Am}$ ", *New Phys.: Sae Mulli*, **vol.72**, 173 (2022) [doi:10.3938/NPSM.72.173]
3. Sungwoon Choi, **Minjae Kwon**, In-Kwon Yoo, "Study of the Fake Hit Effect Observed During the Charge Threshold Scan Procedure of the ALICE Pixel Detector", *New Phys.: Sae Mulli*, **vol.73**, 693 (2023) (*Co-Author*) [doi:10.3938/NPSM.73.693]
4. Saeyoung Jung, **Minjae Kwon**, Hyugyu Choi, Yechan Kim, Inkyu Park, "Multi-physics FEA simulation of residual strain and stress limits in laminated glass for lunar environments", *Asia Pacific International Symposium On Aerospace Technology (APISAT 2024)*, 1685-1693 (2024) (*Co-Author, Conference Paper*) [hyperlink]
5. WooJong Kim, Jason Sang Hun Lee, Inkyu Park, Donghyun Song, Myeonghun Choi, Woohyeon Heo, Donghyun Kim, **Minjae Kwon**, Hyupwoo Lee, "Study on structure of boron-based GEM detector for neutron with Geant4", *New Phys.: Sae Mulli*, **vol.75**, 536 (2025) (*Co-Author*) [doi:10.3938/NPSM.75.536]

6. Yongjun Choi, Kyungrim Woo, In-Kwon Yoo, **Minjae Kwon**, "Study on the cluster size in silicon monolithic active pixel sensors with low energy α -particles", *New Phys.: Sae Mulli*, **vol.76**, 125-132 (2026) (Co-Author) [doi:10.3938/NPSM.76.125]
7. ALICE ITS Collaboration, "Sensor operating point calibration and monitoring of the ALICE Inner Tracking System during LHC Run 3", *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, **vol.1079**, 170596 (2026) [doi:10.1016/j.nima.2026.171354]
8. WooJong Kim, DongHyun Kim, **Minjae Kwon**, Jason Sang Hun Lee, Hyupwoo Lee, Inkyu Park, Donghyun Song, Inseok Yoon, Myeonghun Choi, "Cold Neutron Imaging and Efficiency Measurements with a Boron-10 Coated Double-GEM Detector", *arXiv preprint*, (Co-Author) [arXiv:2602.19468; physics.ins-det]

Languages

Korean	C2: Mother Tongue	English	B2
Japanese	A1	French	A1

Computing Skills

Bold: Well Skilled

Programming Language	C++ , Python , Fortran, Shell Script, Typescript	Monitoring System	Grafana, Prometheus
Mathematics Tool	MATLAB , Mathematica	Physics Package	ROOT , GEANT4
Operating System	Linux , MacOS	Graphical Design	Photoshop , Illustrator , Indesign
Mechanical Design	AutoCAD, Fusion360	Electronic Design	Synopsys CC DC ICC2, KiCAD
Others	LaTeX , react.js, next.js, Docker, HTCondor, Jekyll		